

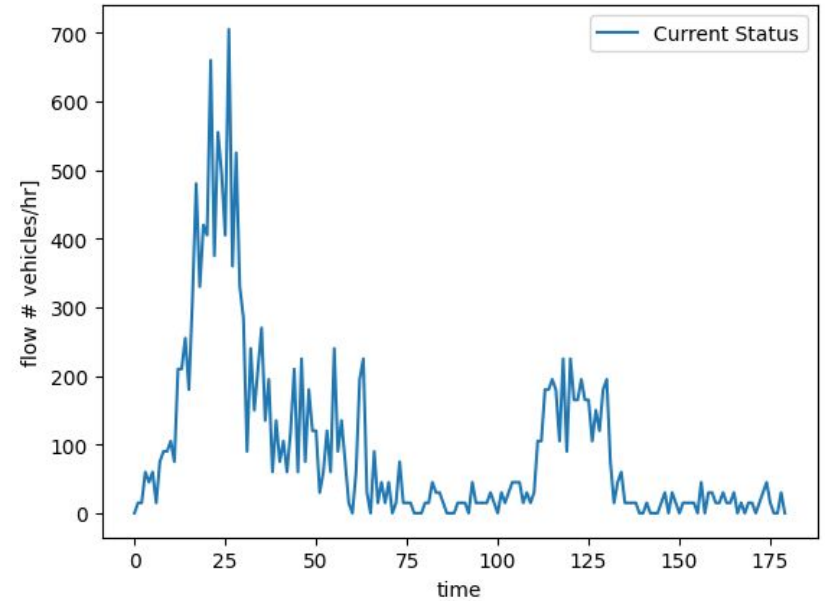
Coco Project 2025

Case study to regulate traffic jam

CoCo P-Control

Summary

- Fast decay of flow
- Grid lock
- High waiting time



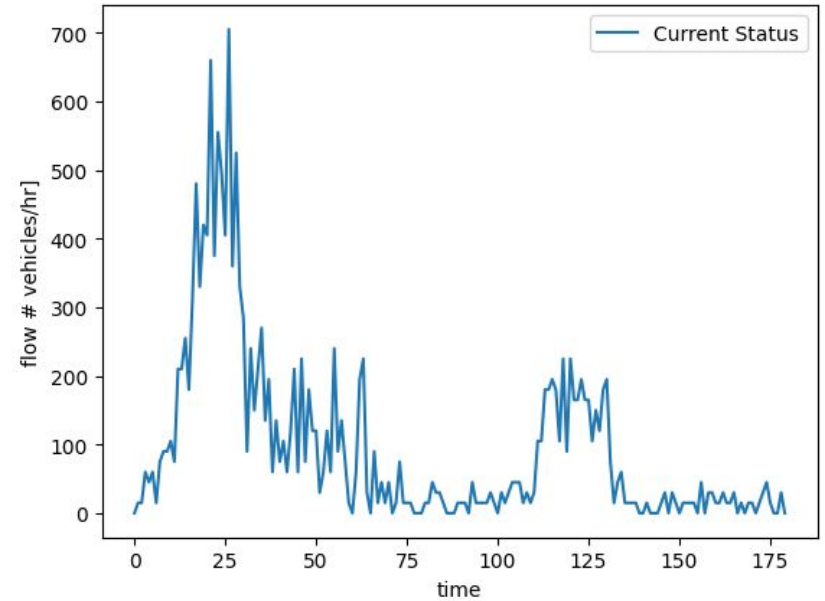
	Current Status
C02_abs	2469.28
travel_time	26.25
waiting_time	25.09
nvehicles	1607.0

CoCo P-Control

Failures

- No prediction forward
- Unstable to changes
- People want better traffic flow

→ use MPC

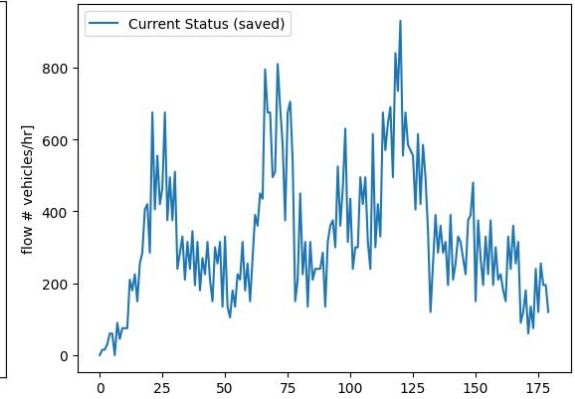
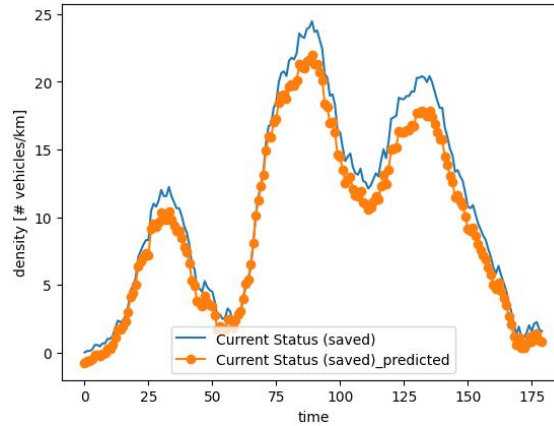


	Current Status
C02_abs	2469.28
travel_time	26.25
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MPC-Control

Suggestions

- Use of MPC Controller
- Reacts on changes and on disturbances
- Fits to model
- Fast in computation



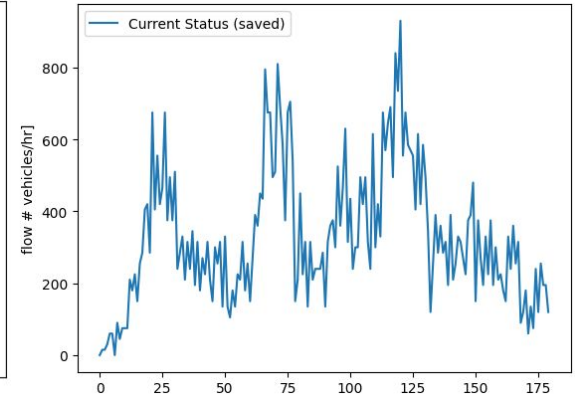
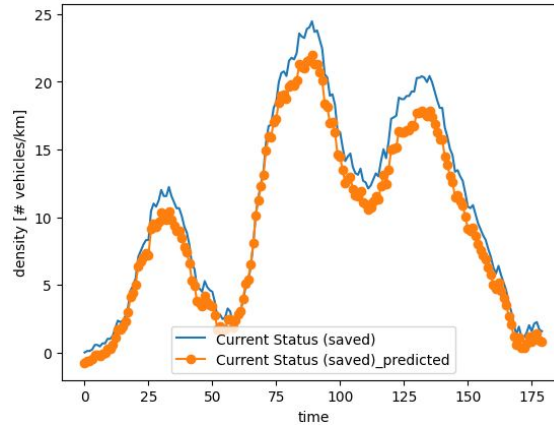
Works without future predictions of spawn!

	Current Status (saved)
C02_abs	743.17
travel_time	6.7
waiting_time	3.77
nvehicles	1703.0

MPC-Control

Deployment

- More Data
- Adaptability to changes in System



Works without future predictions of spawn!

	Current Status (saved)
C02_abs	743.17
travel_time	6.7
waiting_time	3.77
nvehicles	1703.0